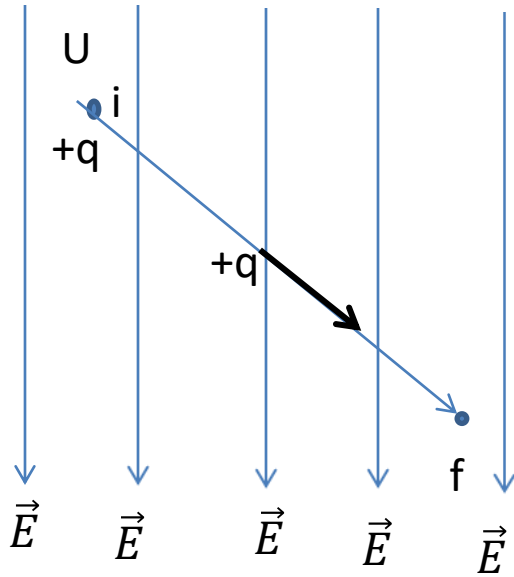




LESSON 4

- BOOK CHAPTER 24
- ELECTRIC POTENTIAL

Electric Potential



The potential energy (U) per unit charge(q) at a point in an electric field ($\vec{E}$) is called the **electric potential V** .

$$V = \frac{U}{q} \dots\dots\dots (i)$$

The electric potential difference between any two points i and f in an electric field is equal to the difference in potential energy per unit charge between the two points:

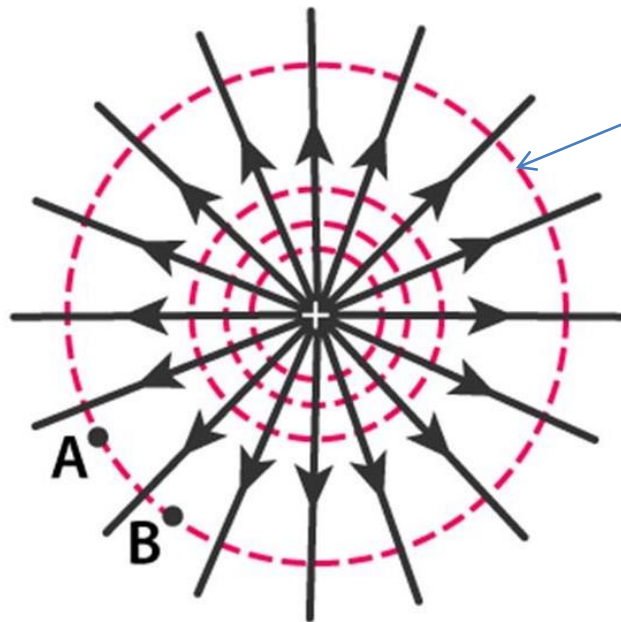
$$\Delta V = V_f - V_i = \frac{U_f}{q} - \frac{U_i}{q} = \frac{\Delta U}{q} \dots\dots\dots (ii)$$

we can define the potential difference between points i and f as

$$\Delta V = V_f - V_i = \frac{-W}{q} \dots\dots\dots (iii)$$

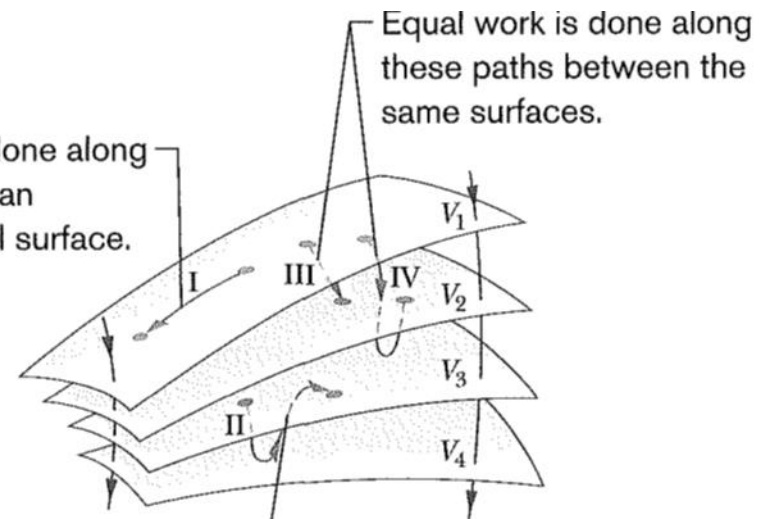
Equipotential Surfaces:

Adjacent points that have the same electric potential form an **equipotential** surface, which can be either an imaginary surface or a real, physical surface. No net work W is done on a charged particle by an electric field when the particle moves between two points A and B on the same equipotential surface.



Equipotential surface

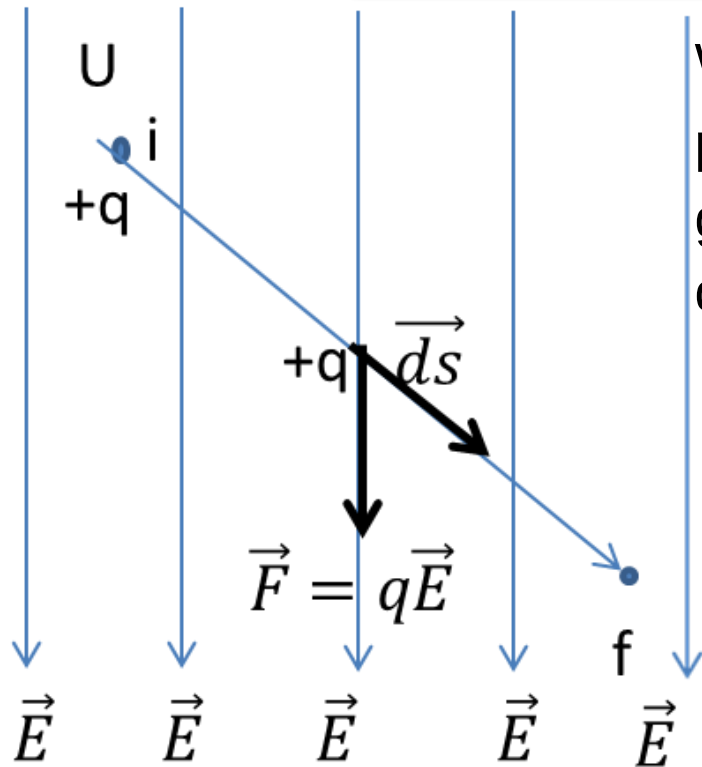
No work is done along this path on an equipotential surface.



Equal work is done along these paths between the same surfaces.

No work is done along this path that returns to the same surface.

Calculating the Potential from the Field



We know that the differential work dW done on a particle by a force $\vec{F}$ during a displacement $d\vec{s}$ is given by the dot product of the force and the displacement:

$$dW = \vec{F} \cdot d\vec{s} = q\vec{E} \cdot d\vec{s}$$

The total work W done on the particle by the field as the particle moves from point i to point f ,

$$W = \int_i^f dW = q \int_i^f \vec{E} \cdot d\vec{s}$$

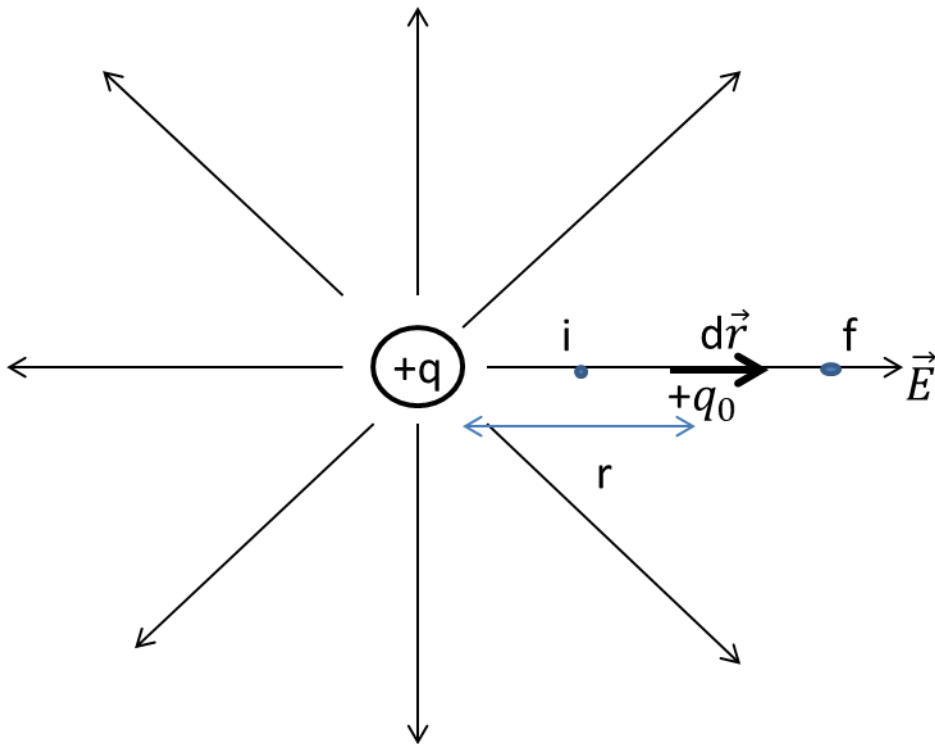
The work done by the electrostatic force in terms of potential difference:

$$\Delta V = V_f - V_i = \frac{-W}{q}$$

$$V_f - V_i = \frac{-q \int_i^f \vec{E} \cdot d\vec{s}}{q} = - \int_i^f \vec{E} \cdot d\vec{s} \quad \text{[Using } W = \int_i^f \vec{E} \cdot d\vec{s} \text{]}$$

Finally, $V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s} \dots\dots\dots \text{(iv)}$

Potential Due to a Point Charge:



We have $V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}$ (iv)

$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{r} \quad [\text{here } d\vec{s} = d\vec{r}]$$

$$V_f - V_i = - \int_i^f E \cos 0^\circ dr$$

$$V_f - V_i = - \int_i^f E dr$$

$$V_f - V_i = - \frac{q}{4\pi\epsilon_0} \int_i^f \frac{1}{r^2} dr$$

$$V_f - V_i = - \frac{q}{4\pi\epsilon_0} \left(- \left| \frac{1}{r} \right|_{r=r_i}^{r=r_f} \right)$$

$$V_f - V_i = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_f} - \frac{1}{r_i} \right]$$

If r_f goes to infinity (∞), then $V_f = 0$ and $\frac{1}{r_f} = 0$

Hence,

$$0 - V_i = \frac{q}{4\pi\epsilon_0} \left[0 - \frac{1}{r_i} \right]$$

$$V_i = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r_i} \right]$$

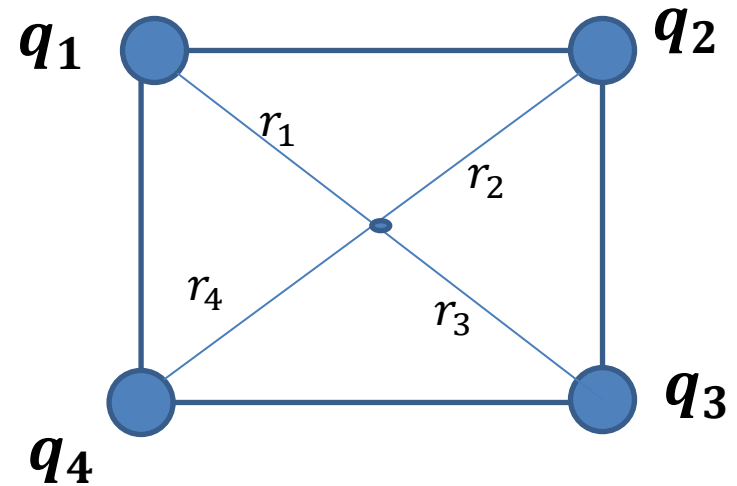
Finally we get,

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$$

Potential Due to a Group of Point Charges:

General Formula:

$$V = \sum_{i=1}^n V_i = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}$$



For an example:
$$V = \sum_{i=1}^4 V_i = \frac{1}{4\pi\epsilon_0} \left[\frac{q_1}{r_1} + \frac{q_2}{r_2} + \frac{q_3}{r_3} + \frac{q_4}{r_4} \right]$$

Problem: 4 (Book chapter 24)

Two large, parallel, conducting plates are 12 cm apart and have charges of equal magnitude and opposite sign on their facing surfaces. An electric force of $3.9 \times 10^{-15} \text{ N}$ acts on an electron placed anywhere between the two plates. (Neglect fringing.) (a) Find the electric field at the position of the electron. (b) What is the potential difference between the plates?

Answer:

Given

$$d = 12 \text{ cm} = 0.12 \text{ m}$$

$$|F| = 3.9 \times 10^{-15} \text{ N}$$

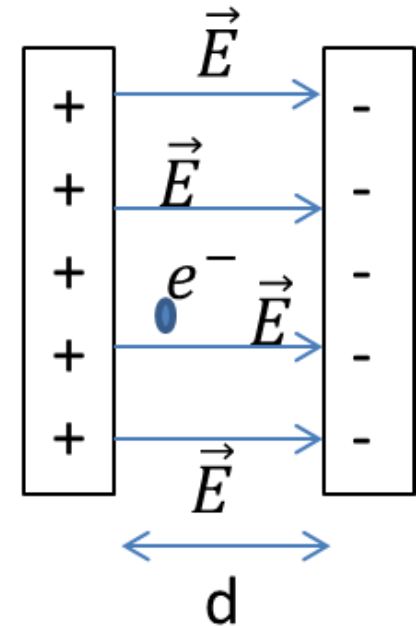
$$|e^-| = |q| = 1.6 \times 10^{-19} \text{ C} \quad |E| = ?$$

(a) We know $|F| = qE$

$$E = \frac{3.9 \times 10^{-15}}{1.6 \times 10^{-19}} = 2.437 \times 10^4 \frac{\text{N}}{\text{C}}$$

(b) The potential difference between the plates

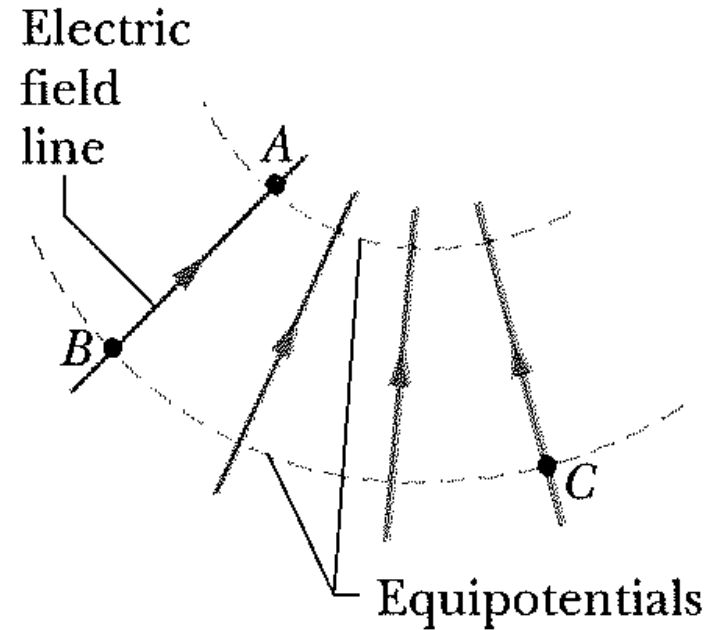
$$|\Delta V| = Ed = (2.437 \times 10^4)(0.12) = 0.293 \times 10^4 \text{ Volt}$$



Problem: 6 (Book chapter 24)

When an electron moves from A to B along an electric field line in the adjacent figure, the electric field does $3.94 \times 10^{-19} \text{ J}$ of work on it. What are the electric potential differences

(a) $V_B - V_A$, (b) $V_C - V_A$, and (c) $V_C - V_B$?



Answer:

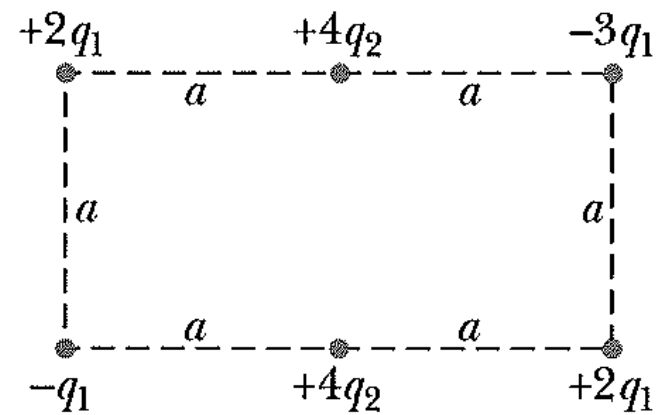
$$(a) \quad V_B - V_A = \frac{-W}{q} = \frac{-3.94 \times 10^{-19}}{-1.6 \times 10^{-19}} = 2.45 \text{ Volt}$$

$$(b) \quad V_C - V_A = 2.45 \text{ Volt} \quad \text{[Because points B and C are on the same equipotential surface]}$$

$$(c) \quad V_C - V_B = 0 \text{ Volt}$$

Problem: 16 (Book chapter 24)

Figure shows a rectangular array of charged particles fixed in place, with distance $a = 39 \text{ cm}$ and the charges shown as integer multiples of $q_1 = 3.4 \text{ pC}$ and $q_2 = 6.0 \text{ pC}$. With $V = 0$ at infinity, what is the net electric potential at the rectangle's center? (*Hint: Thoughtful examination of the arrangement can reduce the calculation.*)



Answer:

Given $a = 39 \text{ cm} = 0.39 \text{ m}$

$$q_1 = 3.40 \text{ pC} = 3.4 \times 10^{-12} \text{ C}$$

$$q_2 = 6.0 \text{ pC} = 6.0 \times 10^{-12} \text{ C}$$

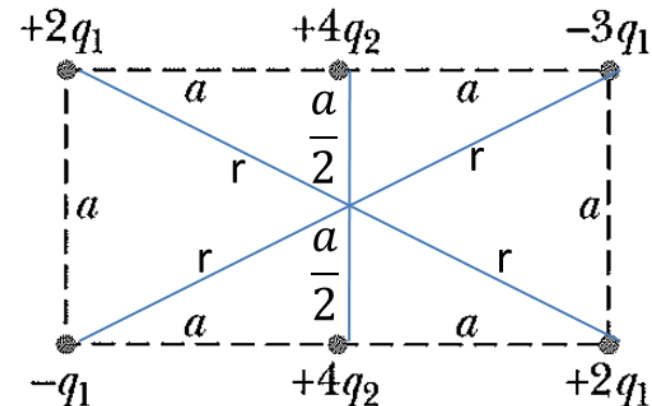
The net potential at the rectangle center is

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{2q_1}{r} + \frac{-3q_1}{r} + \frac{2q_1}{r} + \frac{-q_1}{r} \right] + \frac{1}{4\pi\epsilon_0} \left[\frac{4q_2}{\frac{a}{2}} + \frac{4q_2}{\frac{a}{2}} \right]$$

$$V = 0 + \frac{1}{4\pi\epsilon_0} \left[\frac{4q_2}{0.195} + \frac{4q_2}{0.195} \right]$$

$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{8q_2}{0.195} \right] = \frac{9 \times 10^9 \times 8 \times 6 \times 10^{-12}}{0.195}$$

$$V = 2.215 \text{ Volt}$$



Problem: 17 (Book chapter 24)

In the adjacent Figure, what is the net electric potential at point P due to the four particles if $V = 0$ at infinity, $q = 5 \text{ fC}$, and $d = 4 \text{ cm}$?

Answer:

Given

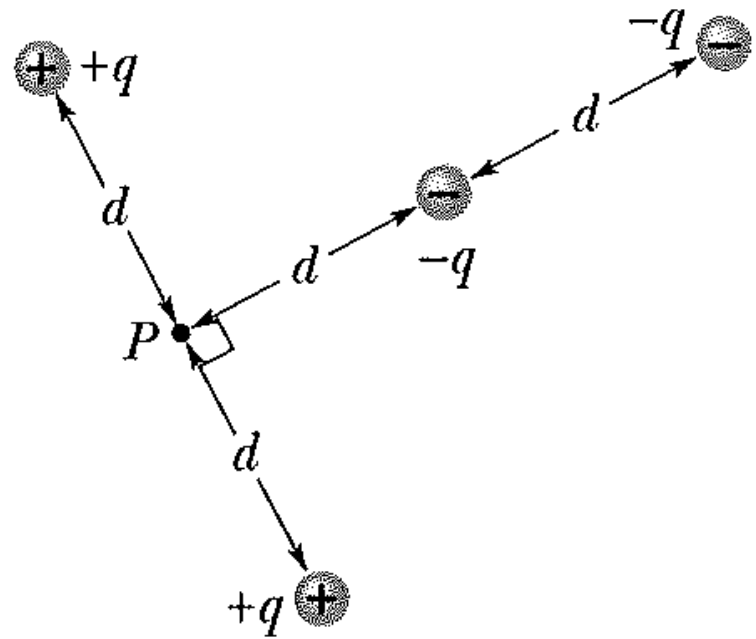
$$q = 5 \text{ fC} = 5 \times 10^{-15} \text{ C} \quad \text{and} \quad d = 4 \text{ cm} = 0.04 \text{ m}$$

The net electric potential at the point P is

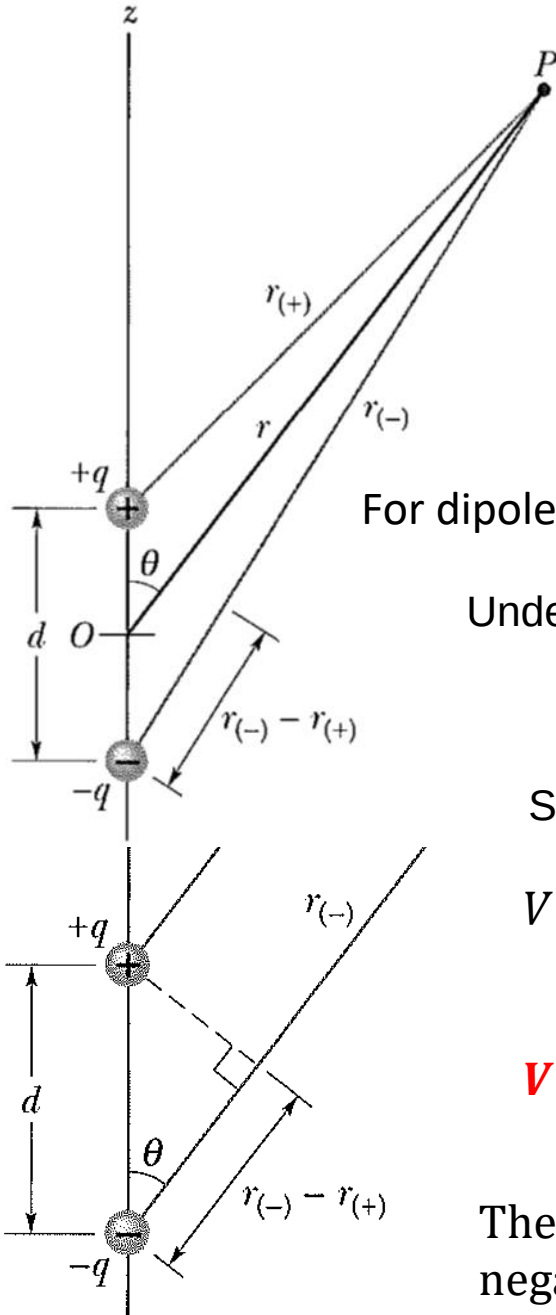
$$V = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{d} + \frac{q}{d} + \frac{-q}{d} + \frac{-q}{2d} \right]$$

$$V = \frac{9 \times 10^9 \times q}{d} \left[1 + 1 - 1 - \frac{1}{2} \right]$$

$$V = \frac{9 \times 10^9 \times 5 \times 10^{-15}}{0.04 \times 2} = 562.5 \times 10^{-6} \text{ Volt}$$



Potential Due to an Electric Dipole:



The net potential at P is given by

$$V = \sum_{i=1}^2 V_i = V_{(+)} + V_{(-)} = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{r_{(+)}} + \frac{-q}{r_{(-)}} \right]$$

$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{r_{(-)} - r_{(+)}}{r_{(-)}r_{(+)}} \right)$$

For dipoles, usually, $r \gg d$, where d is the distance between the charges.

Under the above condition, from the figure we can write

$$r_{(-)} - r_{(+)} = d \cos \theta \quad \text{and} \quad r_{(-)}r_{(+)} \approx r^2$$

Substituting these values in the above equation, we get

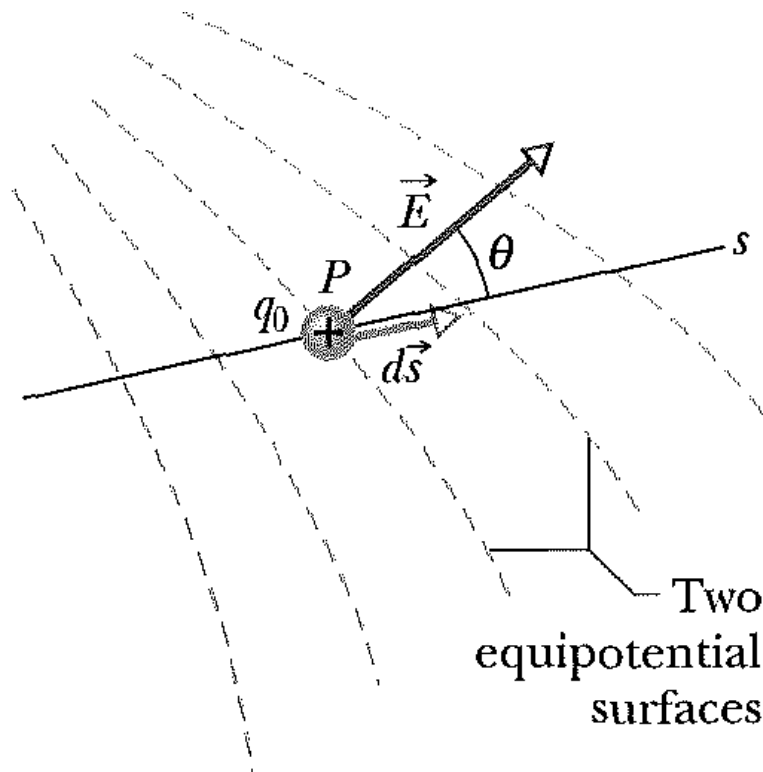
$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{d \cos \theta}{r^2} \right) \quad \text{where, } \theta \text{ is measured from the dipole axis as shown in the figure.}$$

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{p \cos \theta}{r^2} \right)$$

where, $p (= qd)$ is the magnitude of electric dipole moment $\vec{p}$.

The vector $\vec{p}$ is directed along the dipole axis, from the negative to the positive charge.

Calculating the Electric Field from the Electric Potential:



Suppose that a positive test charge q_0 moves through a displacement $d\vec{s}$ from one equipotential surface to the adjacent surface.

The differential work done in terms of electric potential difference dV is

$$dW = -q_0 dV$$

The differential work done by the electric field $\vec{E}$ is

$$dW = \vec{F} \cdot d\vec{s} = q_0 \vec{E} \cdot d\vec{s} = q_0 E (\cos\theta) ds$$

Hence, we get
$$-q_0 dV = q_0 E (\cos\theta) ds$$

$$E (\cos\theta) = -\frac{dV}{ds}$$

Since $E \cos \theta$ is the component of $\vec{E}$ in the direction of $d\vec{s}$, then
$$E_s = -\frac{\partial V}{\partial s}$$

If we take the s axis to be, in turn, the X , y , and z axes, we find that the X , y , and z components of $\vec{E}$ at any point are

$$E_x = -\frac{\partial V}{\partial x} ; \quad E_y = -\frac{\partial V}{\partial y} ; \quad E_z = -\frac{\partial V}{\partial z}$$

THANK YOU